

Adding Wireless Capability to a PRC

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Conference 2000

Overview

- INetLib Intro
- Raw Data Transfer
  - Getting Started
  - The Two Roads
  - Wrapping Up
- Interfacing with Clipper
- Summary



INetLib Intro

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What Is INetLib?

- A library with an API designed to simplify Internet access
  - Integrated Wireless or via a peripheral
  - HTTP and HTTPS
- Network layer over NetLib

Why Use INetLib?

- For sending/receiving raw data
  - Use Palm™ VII series wireless radio
  - More control than WCAs and Clipper
  - Use HTTP and HTTPS



Raw Data Transfer

Getting Started

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Getting Started

- What's needed
- Files to include
- Check for presence
- Open the library
- Configure as needed
  - Setting the conversion algorithm
  - Example setting
- Event monitoring

What's Needed

- Development environments, e.g.:
  - Metrowerks CodeWarrior® for Palm OS®
  - PRC Tools (GCC)
- SDK
  - 3.2, 3.5, or 4.0
- ROM images
  - Palm™ VII series, Palm OS 4.0
- Palm OS Emulator

Files to Include

- INetMgr.h
  - INetLib API declarations and constants
- CTP.h
  - Clipping proxy server instruction constants
    - E.g., the conversion algorithm to use
- Lz77Mgr.h
  - For Lz77 compression
  - If using 3.5 SDK, need to install Lz77lib.prc with app

Check for Presence

- INetLib must be available

```
// check if INetLib is installed
err = FtrGet(inetLibFtrCreator,
             netFtrNumVersion,
             &value);

// find the reference number
if(!err && value)
  err = SysLibFind("INet.lib", &libRefNum);
if(!err)              // then you're ready
```

Open the Library

- INetLibOpen ()
  - Opens the NetLib and INetLib
    - 33 KB of RAM on top of 32 KB required by system
    - More when using HTTPS and/or a cache
  - Provides an inet handle
    - Unique to your app
  - Specify the configuration
    - Default wireless and “wireline” configuration available, e.g.:

      `inetCfgNameCTPDefault`
  - Retrieves default settings
    - Network or Wireless Preferences panel
      - Depending on the configuration and hardware

Configure as Needed

- INetLibSettingSet ()
  - Affects only the inet handle for the calling app
  - Changes in affect until INetLibClose ()
  - Set:
    - Conversion algorithm
    - The cache (if any) and its maximum size
    - Cookies enabled/disabled
    - A different configuration to use

Setting the Conversion Algorithm

- Use ***inetSettingConvAlgorithm*** constant
- Values
  - ***ctpConvNone***
    - Raw data
  - ***ctpConvNoneLZ77***
    - Raw compressed data on download
    - Use Lz77 API to decompress
    - Supports alphabets not having lower-case characters

Example Setting

- Conversion algorithm setting

```
UInt32 convType = ctpConvNoneLZ77;

// Set conversion algorithm for raw data
err = INetLibSettingSet(
                  libRefNum,
                  inetH,
                  inetSettingConvAlgorithm,
                  (BytePtr) &convType,
                  sizeof(convType)
                           );
```


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Event Monitoring

- INetLibGetEvent ()
  - Replaces EvtGetEvent ()
    - *inetSockReadyEvent*
    - *inetSockStatusChangeEvent*
      - e.g.: connecting, sending, waiting, receiving
    - Everything else like EvtGetEvent ()



Raw Data Transfer

The Two Roads

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The High Road

- INetLibURLOpen ()
  - **GET method only**
  - Pass a URL
    - Protocol is extracted from the URL
  - Equivalent to:
    - INetLibURLCrack ()
    - INetLibSockOpen ()
    - INetLibHTTPReqCreate ()
    - INetLibHTTPReqSend ()

The Low Road

- GET or POST
- More control
- The Calls
  - INetLibURLCrack ()
  - INetLibSockOpen ()
  - INetLibSockSettingSet ()
  - INetLibSockHTTPAttrSet ()
  - INetLibSockHTTPReqCreate ()
  - INetLibSockHTTPReqSend ()

The Low Road

- INetLibSockSettingSet ()
  - Configure the socket
    - *inetSockSettingContentType*
      - For setting the request HTTP Content-type header
      - Prevents URL encoding of data at WCP server
      - Responsibility of the developer to encode data

Example Setting

- Content-type HTTP request setting

```
error = INetLibSockSettingSet(
                                inetRefNum,
                                inetSockH,
                                inetSockSettingContentType,
                                myContTypeStrP,
                                StrLen(myContTypeStrP)
                                );
```

  - Content string ptr MUST NOT be NULL
  - Content-type defaults to "application/octet-stream”
    - When content string is “”

The Low Road

- INetLibSockHTTPAttrSet ()
  - *inetHTTPAttrIncHTTP*
    - Includes HTTP response headers in data when
      - Set to true
      - *ctpConvNone* or *ctpConvNoneLZ77*

The Low Road

- INetLibSockConnect ()
  - Not necessary to invoke
  - Called by INetLibSockHTTPReqSend ()
    - Will use correct settings

The Low Road

- INetLibSockHTTPReqSend ()
  - Method determined here
  - For POST
    - Pass pointer to content data
  - For GET
    - Pass NULL as the data ptr



Raw Data Transfer

Wrapping Up

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Retrieving Data

- INetLibSockRead ()
  - Retrieve and store for each *inetSockReadyEvent*
    - May initially need to wait for several events
    - 0 bytes read **after** data retrieval indicates end of data
  - Blocks until at least 1 byte read
    - Use timeout argument to exit blocking state

Closing Down

- INetLibSockClose ()
  - Closes the socket & frees memory
- INetLibClose ()
  - Closes the inet handle
    - Also closes any open sockets associated with the app
  - Frees memory
  - Open count kept to prevent closure if other apps using INetLib
    - Timer started if open count decrements to 0



Interfacing with Clipper

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Interfacing with Clipper

- SysUIAppSwitch ()
  - 2 launch commands:
    - *sysAppLaunchCmdOpenDB*
      - Pass cardNo and dbID for Clipper
      - Pass a parameter block structure containing
        - » cardNo where WCA resides
        - » dbID of WCA
      - Creates URL in form of file:WCAname.pqa
    - *sysAppLaunchCmdGoToURL*
      - Simply pass a URL, e.g.:

        http://www.example.com

        file:WCAname.pqa



Development Resources

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Development Resources

- www.palmos.com/dev/tech/
  - Documents
    - Programmer's Companion
    - SDK Reference
    - FAQ articles in knowledge base
  - Example code (in KB)
    - INetLow
  - Tools
    - IDEs
    - Palm OS® Emulator
    - ROM images in Provider Pavilion

Development Resources

- Development Zone Knowledge Base
  - oasis.palm.com/dev/kb/
  - Searchable
- Developer Assistance
  - www.palmos.com/dev/tech/support/
  - E-mail forum subscription
  - Newsgroup listings
  - Training info

Conclusion

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Conclusion

- The INetLib API provides easy Internet access
- Must use INetLib for internal Palm™ VII radio
  - Allows for connections through serial port/IR too
- Let Clipper request and render Internet data
  - Avoid coding other than launching Clipper

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collaborate

innovate

communicate

Q&A


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